Style Scout: One Category, One Catalog Visual Search Engine

Project Overview

- **Category:** Fashion-Tech
- **Team Members:** Nyshea, Tageena, Valeria, Selenia

1. Problem Statement

Shoppers frequently encounter clothing items online or in person that they want to purchase but cannot easily describe with keyword queries. Manual text searches across multiple retail websites consume 30 to 45 minutes per outfit discovery session and consistently return irrelevant matches due to ambiguity in cuts, patterns, and style naming conventions.

2. Solution

Style Scout is a scoped visual search MVP focused exclusively on tops (shirts, blouses, sweaters). The workflow enables users to upload an outfit screenshot and crop the specific garment. Using pre-computed FashionCLIP embeddings, the application queries an indexed collection of the open H&M Kaggle catalog (105,542 articles) to return the top 5 visually similar matching items with real-time pricing and direct catalog links.

3. Target Market

The primary demographic consists of Gen Z and Millennial shoppers in the United States who discover fashion via social platforms such as Instagram, TikTok, and Pinterest.

- **Market Validation:** According to eMarketer (2026) and McKinsey (2026), 59% of Gen Z and Millennial consumers report being likely to purchase budget-friendly or secondhand fashion items through visual discovery channels.
- **Addressable Audience:** The total addressable online fashion market in the US exceeds 100M digital shoppers.

4. Why It Is Valuable

Style Scout eliminates the friction of keyword guesswork by linking visual inspiration directly to actionable inventory. By constraining the scope to tops and utilizing a single pre-indexed catalog centered around an accessible price threshold ("Find this look under $40"), the product establishes an accurate, benchmarked top-5 visual search experience within a feasible 4 to 6-week engineering schedule.

5. How You Will Make Money

Style Scout employs a freemium model complemented by affiliate marketing monetization:

- **Subscription Tier:** Users receive 5 free visual queries per day; unlimited searches and advanced multi-category filters are offered at $2.99 per month.
- **Affiliate Referral:** Phase 1 focuses on validating commercial affiliate link conversion, projecting an estimated 4% commission rate on apparel referral checkouts in alignment with standard retail partner terms.

6. MVP Features

- **Image Upload & Cropping Tool:** Client-side interface to upload outfit screenshots and adjust a bounding box targeted specifically at tops.
- **FashionCLIP Vector Search:** Generation of dense visual vector embeddings for user cropped images via pre-trained FashionCLIP, querying an index of the H&M catalog.
- **Top-5 Visual Matches Display:** Clean presentation rendering the top 5 visual matches alongside retail item titles, prices, and direct catalog links.

7. Timeline and Division of Work

Week	Team Member	Primary Deliverables
Week 1	Nyshea	Data preprocessing & catalog indexing (H&M Kaggle dataset, 105.5k tops).

Week	Team Member	Primary Deliverables
	Selenia	Core Flask/FastAPI backend initialization and FashionCLIP model integration via pip.
	Valeria	Frontend image upload interface and HTML5 canvas cropping UI.
	Tageena	GitHub repository configuration, baseline README docs, and competitive research.
Week 2	Tageena	Vector similarity search module implementation (FAISS / Cosine similarity metrics).
	Selenia	RESTful API endpoint connecting frontend cropping output to backend vector search.
	Valeria	Results UI design rendering top 5 match product cards with prices and outbound links.
	Nyshea	Market research compilation and structured usability feedback survey design.
Week 3	Nyshea	Benchmark evaluation (top-5 retrieval accuracy test on 100 sample outfit screenshots).

Week	Team Member	Primary Deliverables
	Selenia	Backend error handling, caching, and query response latency optimization.
	Valeria	Bookmarking/saved items feature and responsive mobile layout optimization.
	Tageena	User testing coordination (conducting 10 target tester feedback sessions).
Week 4	All Team Members	Full end-to-end integration, bug remediation, slide deck preparation, demo rehearsal, and final submission.

8. Team Roles and Responsibilities

Role	Member	Responsibilities	Share of Codebase
Lead Data Engineer	Nyshea	Data pipeline engineering, H&M catalog indexing, FashionCLIP embedding generation, and vector search architecture.	30%
Lead Backend Developer	Selenia	API design, model deployment integration, server-side search routing, and latency optimization.	25%
Lead Frontend Developer	Valeria	UI/UX layout, image upload and canvas crop component,	25%

Role	Member	Responsibilities	Share of Codebase
		match display cards, and bookmarking functionality.	
Developer & Research Lead	Tageena	User testing protocols, competitive analysis synthesis, pitch deck creation, technical documentation, and survey reporting.	20%
Total Allocation			**100%**

9. Viability: How We Will Prove This Works

- **User Testing Plan:** Execution of 10 structured benchmark testing sessions with target demographic participants across 100 real-world outfit screenshots, evaluating match quality and system responsiveness.
- **Viability Metric:** Target a verified top-5 match accuracy rate (defined as the user confirming that at least 1 of the top 5 returned items is visually similar to the source query) across 100 real outfit test cases, reported as a clean percentage metric.
- **Competitive Analysis (Checked October 2026):** Generalist visual tools such as Google Lens and Pinterest Visual Search index broad, non-specific cross-category inventory without targeted budget constraints. Style Scout differentiates by focusing exclusively on apparel tops backed by scoped, budget-aligned indexing ("under $40").

10. Scalability: Roadmap from MVP to 100K Users

Phase	Timeline	Technical & Operational Objectives	Target Scale & Revenue Model
Phase 1	Semester (4–6 Weeks)	Scoped web MVP focusing strictly on tops using a local pre-indexed H&M catalog (105,542 articles).	100 test users; Free pilot evaluation.
Phase 2	Months 3–6	Catalog expansion across dresses, pants, outerwear, and footwear using multi-retailer affiliate feeds; cloud vector database migration (Pinecone/Qdrant).	10,000 active users; $2.99/mo premium tier + 4% apparel affiliate commissions.
Phase 3	Months 6–12	Native mobile applications (iOS/Android), real-time camera search, and automated background isolation.	100,000 users; Enterprise retail search API licensing and affiliate monetization.

11. Sources and References

- H&M Personalized Fashion Recommendations Dataset (Kaggle dataset: 105,542 articles with images and metadata).
- FashionCLIP Model Paper & PyPI Package (Patrick John Chia et al., FashionCLIP embedding framework).
- McKinsey & Company / BoF Fashion Outlook Report (2026): Consumer shopping habits and budget/secondhand fashion trends (59% Gen Z/Millennial likelihood).
- eMarketer (2026): US Social Commerce and Visual Shopping Research Data.